

## Bacillus clausii for Gastrointestinal Disorders: A Narrative Literature Review.

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The gut microbiota is intrinsically linked to human health; disturbances in microbial homeostasis are implicated in both intestinal and extraintestinal disorders. Probiotics are "live microorganisms that, when administered in adequate amounts, confer a health benefit on the host," and many commercial preparations comprising a diverse range of species are available. While probiotics have been much researched, better understanding of the probiotic effects and applications of species such as *Bacillus clausii* is warranted. In this narrative literature review, we review the characteristics and mechanisms of action supporting *B. clausii* as a probiotic and discuss the evidence from clinical studies evaluating *B. clausii* probiotics for the management of a variety of gastrointestinal disorders and symptoms in children and adults. Finally, we highlight the challenges of future research and the need for more robust and diverse clinical evidence to guide physicians in the clinical application of probiotics for gastrointestinal disorders and other conditions.

## Composite genome sequence of Bacillus clausii, a probiotic commercially available as Enterogermina®, and insights into its probiotic properties.

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Some of the spore-forming strains of *Bacillus* probiotics are marketed commercially as they survive harsh gastrointestinal conditions and bestow health benefits to the host. We report the composite genome of *Bacillus clausii* ENTPro from a commercially available probiotic Enterogermina® and compare it with the genomes of other *Bacillus* probiotics. We find that the members of *B. clausii* species harbor high heterogeneity at the species as well as genus level. The genes conferring resistance to chloramphenicol, streptomycin, rifampicin, and tetracycline in the *B. clausii* ENTPro strain could be identified. The genes coding for the bacteriocin gallidermin, which prevents biofilm formation in the pathogens *Staphylococcus aureus* and *S. epidermidis*, were also identified. KEGG Pathway analysis suggested that the folate biosynthesis pathway, which depicts one of the important roles of probiotics in the host, is conserved completely in *B. subtilis* and minimally in *B. clausii* and other probiotics. We identified various antibiotic resistance, bacteriocins, stress-related, and adhesion-related domains, and industrially-relevant pathways, in the genomes of these probiotic bacteria that are likely to help them survive in the harsh gastrointestinal tract, facilitating adhesion to host epithelial cells, persistence during antibiotic treatment and combating bacterial infections.

## Quantitative and qualitative characterization of commercially available oral suspension of probiotic products containing Bacillus Clausii spores.

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Probiotics contain beneficial live bacteria that confer several health benefits to the host. For the past 50 years, spore-forming *Bacillus* species have been used in the form of probiotics. Among these, *Bacillus clausii* strains are used for the management of acute and antibiotic-associated diarrhoea. In the present work, we have evaluated the asserted label information on randomly chosen commercial *Bacillus clausii* spore suspension of probiotic products. The quality and number of viable bacteria were evaluated based

on the colony count, antibiotic resistance, and hemolytic activity assays. The colony fingerprinting and 16S rRNA gene-sequencing techniques were used to confirm the presence of a univariate strain (*Bacillus clausii*). Our results corroborated the label count of  $2 \times 10^9$  CFU/5 mL in BACIPRO®, ENTEROGERMINA®, and TUFPRO® products. However, vegetative spore count was not found to match with the given label count in BENEGUT®, PROALANA-B®, β-LOCK®, and PROCILLUS® *Bacillus clausii* brands. In the hemolytic activity assay, except for β-LOCK®, the other 6 products showed gamma-hemolysis activity. *Bacillus clausii* isolated from all 7 probiotic products demonstrated resistance to several broad-spectrum antibiotics. The 16S rRNA gene-sequencing data detected genera of *Bacillus* and *Bacillus clausii* strain in the BACIPRO®, ENTEROGERMINA®, PROALANA-B®, BENEGUT®, and TUFPRO® products; however, *Ralstonia mannitolilytica* and *Paenibacillus dendritiformis* species were identified in β-LOCK® and PROCILLUS®, respectively. As correct label information was observed only in BACIPRO®, ENTEROGERMINA®, and TUFPRO® products, it is proposed that a more stringent quality check would minimize the possibility of mismatch concerning the label information.

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## Bacillus strains as human probiotics: characterization, safety, microbiome, and probiotic carrier.

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Both spore and vegetative forms of *Bacillus* species have been used as probiotics, and they have high stability to the surrounding atmospheric conditions such as heat, gastric conditions, and moisture. The commercial *Bacillus* probiotic strains in use are *B. cereus*, *B. clausii*, *B. coagulans*, *B. licheniformis*, *B. polyfermenticus*, *B. pumilus*, and *B. subtilis*. These strains have antimicrobial, anticancer, antioxidant, and vitamin production properties. However, *Bacillus* probiotics can also produce toxins and biogenic amines and transfer antibiotic resistance genes; therefore, their safety is a concern. Studies on the microbiome using probiotic *Bacillus* strains are limited in humans. Most microbiome research has been conducted in chicken, mouse, and pig. Some *Bacillus* probiotics are used as fermentation starters in plant and soybean and dietary supplement of baking foods as a probiotic carrier. This review summarizes the characterization of *Bacillus* species as probiotics for human use and their safety, microbiome, and probiotic carrier.

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## A tale of caution: prolonged *Bacillus clausii* bacteraemia after probiotic use in an immunocompetent child.

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*Bacillus clausii* as a probiotic supplement is increasingly used in both adult and paediatric patient populations. There is limited awareness about potential adverse effects. We report a case of prolonged (111 days) *B. clausii* bacteraemia after brief probiotic use in a 17-month-old immunocompetent child, without a definite focus of infection and in the absence of predisposing risk factors or underlying co-morbidities. We identified seven probiotic use-associated cases of prolonged *B. clausii* bacteraemia (mean duration [range] 64 days [14-93 days] where data were available) in the literature, all with underlying co-morbidities. *B. clausii* probiotic preparations may cause prolonged bacteraemia, rendering patients with underlying co-morbidities as well as those with unrecognized risk factors vulnerable for significant infectious complications.

## Role of a probiotic strain in the modulation of gut microbiota and cytokines in inflammatory bowel disease.

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To assess the effect of a probiotic strain *Bacillus clausii* UBBC-07 on gut microbiota and cytokines in IBD patients. Patients were randomly allocated to either placebo or probiotic *Bacillus clausii* UBBC-07 for four weeks along with the standard medical treatment (SMT). Enrolled patients were evaluated before and after intervention for presence of the given probiotic, change in gut microbiota, change in serum cytokines, serotonin and dopamine, symptoms of disease, physical, behavioral and psychological parameters. Probiotic strain *Bacillus clausii* UBBC-07 showed good survival in IBD patients in the treatment group ( $p < 0.01$ ) without any reported adverse event. Metagenomic analysis showed that the given probiotic strain was able to modulate the gut microbiota in treated group. Phylum Firmicutes was increased and phylum Bacteroidetes was decreased in the probiotic treated group. A significant increase was observed in the abundance of anaerobic bacterial genera *Lactobacillus*, *Bifidobacterium* and *Faecalibacterium* in the probiotic treated group ( $p < 0.01$ ) as compared to placebo group. Significant increase was observed in IL-10 ( $p < 0.05$ ) and variable decrease in the secretion of IL-1 $\beta$ , TNF- $\alpha$ , IL-6, IL-17 and IL-23 in probiotic treated group. In the treatment group a significant decrease in the symptoms of IBD and improvement in the psychological parameter to various degrees was noted. These results indicated that probiotic strain *B. clausii* UBBC-07 affected the gut microbiota and cytokine secretion and shown efficacy in IBD patients.

## Protective action of *Bacillus clausii* probiotic strains in an in vitro model of Rotavirus infection.

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Rotavirus is the most common cause of acute gastroenteritis (AGE) in young children. *Bacillus clausii* (*B. clausii*) is a spore-forming probiotic that is able to colonize the gut. A mixture of four *B. clausii* strains (O/C, T, SIN and N/R) is commonly used for the treatment of AGE, and it has been demonstrated that it can reduce the duration and severity of diarrhea in children with AGE. Few studies have sought to characterize the mechanisms responsible for such beneficial effects. Intestinal effects of probiotics are likely to be strain-specific. We conducted a series of in vitro experiments investigating the activities of this mixture of *B. clausii* strains on biomarkers of mucosal barrier integrity and immune function in a cellular model of Rotavirus infection. *B. clausii* protected enterocytes against Rotavirus-induced decrease in trans-epithelial electrical resistance, and up-regulated expression of mucin 5AC and tight junction proteins (occludin and zonula occludens-1), all of which are important for effective mucosal barrier function. *B. clausii* also inhibited reactive oxygen species production and release of pro-inflammatory cytokines (interleukin-8 and interferon- $\beta$ ) in Rotavirus-infected cells, and down-regulated pro-inflammatory Toll-like receptor 3 pathway gene expression. Such mechanisms likely contributed to the observed protective effects of *B. clausii* against reduced cell proliferation and increased apoptosis in Rotavirus-infected enterocytes.

## Bacillus clausii Attenuates 5-Fluorouracil-Induced Intestinal Mucositis in Mice.

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5-Fluorouracil (5-FU) is an antimetabolite widely prescribed in cancer treatments, but its use in highly proliferative tissues can cause significant problems such as mucositis. *Bacillus clausii* is a probiotic commonly used for protection against acute diarrhea, gastrointestinal dysbiosis and inflammatory bowel diseases. We investigated the effect of *B. clausii* on 5-FU intestinal mucositis in mice. *B. clausii* was administered concomitantly with 5-FU on the first day and alone for the other two days. After the third day of 5-FU (450 mg/kg, ip), the animals were euthanized. Ileum samples were removed for evaluation for histopathological and biochemical analyses (myeloperoxidase (MPO), glutathione (GSH), malondialdehyde (MDA), catalase (CAT), interleukin-1 beta (IL-1 $\beta$ ) and tumor necrosis factor alpha (TNF- $\alpha$ ). In addition, we investigated gastric emptying and intestinal transit, intestinal permeability, intestinal smooth muscle contractility, transepithelial electrical resistance and intestinal transport of water and electrolytes. *B. clausii* reduced histopathological scores and increased the villus/crypt ratio in all intestinal segments after mucositis. *B. clausii* attenuated 5-FU-induced weight loss. The probiotic treatment increased GSH levels, reduced MPO and CAT activity, and also reduced MDA, IL-1 $\beta$  and TNF- $\alpha$  levels. *B. clausii* attenuated the delay in gastric emptying, water and electrolyte secretion and intestinal hypercontractility, and increased 5-FU-induced intestinal permeability. Thus, our data suggest that *B. clausii* may be a potential candidate for the treatment of chemotherapy-induced intestinal mucositis.

## Survival and Germination of Bacillus clausii UBBC07 Spores in in vitro Human Gastrointestinal Tract Simulation Model and Evaluation of Clausin Production.

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*Bacillus clausii* UBBC07 is a commercial spore probiotic known to reduce diarrhea in children and adults. In the present study, survival and germination of UBBC07 spores were investigated under fed and fasted conditions in Simulator of Human Intestinal Microbial Ecosystem. Besides this, lantibiotic production, purification, and characterization were performed. The agar plate analysis showed that spores were 100% tolerant to fed and fasted gastrointestinal tract (GIT) conditions. Simultaneously, flow cytometry revealed that at the end of small intestinal incubation, 120% (fed) and 133% (fasted) spores were in viable germinating state. The transformation of viable germinating spores into viable vegetative cells was observed at 3 h of incubation under fasted GIT conditions. In antimicrobial evaluation, UBBC07 produced low-molecular-weight (2107.94 Da) class I lantibiotic clausin. The presence of *lanB*, *lanC*, and *lanD* genes confirms the clausin production. Clausin is stable at proteases (pepsin, proteinase K, and trypsin), temperature (up to 100°C), and pH (up to 11). Furthermore, the antimicrobial activity toward Gram-positive bacteria including *Clostridium difficile* is advantageous. In conclusion, *B. clausii* UBBC07 spore probiotic is capable of surviving and germinating under in vitro upper GIT conditions. The clausin production justifies strain applicability in diarrhea.

## On the fate of ingested Bacillus spores.

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Spores of various *Bacillus* species, including *B. subtilis*, *B. cereus* and *B. clausii*, are used as probiotics,

although they are generally absent from the normal microflora of man. We used two nonpathogenic *Bacillus* species, *B. subtilis* and *B. clausii*, to follow the fate of spores inoculated intragastrically in mice. We did not find detectable amounts of vegetative cells in intestinal samples, probably because of high toxicity of the conjugated bile salt taurodeoxycholic acid against *Bacillus* species. Both spores and cells were detected in the lymph nodes and spleen of one mouse. Our results indicate that *Bacillus* is present in the intestinal tract solely as spores and that nonpathogenic *Bacillus* spores may germinate in lymphoid organs, a finding reminiscent of *B. anthracis* germination in macrophages. These results indicate that any claimed probiotic effect of *B. subtilis* should be due to spores or, alternatively, to vegetative growth outside the intestine.

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